A-A-53367A
19 June 1989
SUPERSEDING
A-A-53367
22 February 1988

COMMERCIAL ITEM DESCRIPTION

SONIC PROPHYLAXIS UNIT, DENTAL, Portable, Air Driven

The General Services Administration has authorized the use of this Commercial Item Description.

This Commercial Item Description covers an air driven, sonic prophylaxis unit suitable for use in removing dental calculus.

Salient Characteristics:

Shall be an air driven (non-electric), sonic prophylaxis unit suitable for use in removing dental calculus. The sonic prophylaxis unit shall consist of one handpiece, swivel assembly, three tip assemblies, and accessories as specified herein.

The tip assemblies shall consist of:

One (1) "universal" tip suitable for all surfaces for removal of both heavy and medium calculus.

One (1) "perio" tip suitable for lingual, supragingival scaling of mandibular incisors and deep local pockets.

One (1) "sickle" tip suitable for use on sensitive patients and provide alternative access for difficult posterior areas.

Each tip assembly shall be furnished with a pliable, clear or translucent cover for protection and storage. The cover shall be capable of being cold sterilized.

Accessories shall include:

One air pressure gauge.
One round tip wrench.
One package containing one, 2 ounce, aerosol lubricant and conditioner.
One spare parts container containing one rotor, one clean-out wire, four O-rings one plastic clamp wrench, one metal wrench and one elastomeric seal.

AMSC N/A FSC 6520

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Material. The component materials shall be as specified in Table I.

Table I. Component materials.

Handpiece components	
Sheath (front portion)	Black anodized aluminum alloy (6061-T6).
Sheath (proximal end portion)	Martensitic stainless steel, Type 416.
Rear seal	Buna-N.
Front suspension	Buna-N elastomer, "O"-Rings.
Water tube	Stainless steel.
Rotor	Martensitic Stainless steel, Type 410.
Swivel	Black anodized aluminum alloy (6061-T6).
Tip assemblies	
Operative end	Martensitic stainless steel, Type 440C.
Holder	Anodized aluminum alloy (6061-T6).
Base	Martensitic stainless steel, Type 416.
Clean-out wire	Austenitic stainless steel.

Style, design, and dimensions. The unit shall be an air driven, sonic handpiece with removable and interchangeable tip assemblies and a swivel assembly to provide a 360° rotation of the handpiece. The tip assemblies (tip and tip holder) shall properly screw into the handpiece. The swivel shall be held into the handpiece by means of a snap ring. The style, design, and dimensions of the unit shall conform to Defense Personnel Support Center (DPSC) Drawing #23401.

When all components are assembled, the tip assembly (tip and tip holder) and the hollow shaft which extends the length of the handpiece shall form a rigid connection. The shaft shall be suitably suspended (front and rear). The unit shall operate from compressed air which shall drive (rotate) a thin walled rotor by passing through nozzles in the shaft wall causing shaft vibration which is amplified by the tip. The motion of the rotor relative to the shaft shall be eccentric.

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The unit shall be suitable for operation by 1.50, plus 0.07, minus 0.16, standard cubic feet per minute (SCFM) of air at a pressure of 40, plus 0, minus 5 pounds per square inch gage (PSIG) and 25, plus or minus 5, cc/min of water at a minimum pressure of 15 PSIG.

The tips shall be self-tuning and shall vibrate (resonate) at a frequency of 6100 plus or minus 300 Hz with the universal tip amplitude of 0.010 inch, plus 0.001 inch, minus 0.003 inch, peak to peak, orbital motion.

Water shall be delivered to the tip by passing through a water tube in the shaft and then through a small integral tube within the tip holder. The design and vibration characteristics of the tips shall allow water to flow by surface contact effect to within 1/4 inch of the free end of the tip before spraying occurs.

The water channel of the tip assemblies (tips and tip holders) shall be "office cleanable" with the use of clean-out wires to clear any blockage of the water channel. The tip assemblies shall be autoclavable, dry heat sterilizable, and quickly changeable. The tips shall have a diamond cross-section and shall incorporate a spherical tip to give clinical function with minimum tooth damage and maximum patient comfort.

The swivel assembly shall have a 4 hole back end suitable for attachment to a standard, 4-line handpiece tubing assembly, and shall meet ISO standards for dimensional configuration.

The pressure gage shall be suitable for connection between the handpiece and the hose (tubing) end to permit setting and adjustment of air pressure. The gage shall have a range of zero to sixty pounds per square inch (psig) with increments of at least 2 psig (scale shall also be marked with metric graduations). The pressure gage shall be required only for initial adjustments and shall not be required during use of the handpiece.

The weight of the unit (handpiece with a tip connected) shall be 2 ounces plus or minus 1/4 ounce.

Hardness. After proper stress relieving, the Rockwell hardness of the operative ends shall be C56 to C60, when tested in accordance with ASTM E18 or ASTM E92 and E140.

Passivation. All stainless steel shall be properly passivated.

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Corrosion resistance. The tips shall show no signs of corrosion when tested in accordance with the copper sulfate and boil test requirements of ASTM F1089.

Autoclave test. The handpiece and tip assemblies (tip and tip holder), shall be capable of being autoclaved as specified herein without showing any signs of corrosion, reduction in performance, or malfunction.

Operational test. Each handpiece with a tip assembly attached shall be subjected to an operational "run-in" test as specified herein without malfunction or reduction in performance.

Finish. The front portion of the sheath shall have a black anodized finish. The proximal end of the sheath and the swivel shall have a satin or matte finish. The operative ends of the tips shall have a smooth bright finish except for the sharpened edges which shall have a finely honed finish. The operative ends shall have sharp cutting edges and shall be free of defects such as coarse grind marks, cracks, gouges, burrs, feathers, or other imperfections which affect performance.

Instructions. Instructions for installation, use, sterilization, maintenance and repairs, trouble shooting guide, and list of spare parts and accessories shall be provided with each unit.

Identification. The prophylaxis unit shall be marked with the name or registered trademark of the manufacturer. If unit is of foreign manufacture, the unit shall be marked with the country of origin.

Color coding. The holder of each tip assembly shall be color coded as follows: Universal-black, perio-natural and sickel-blue.

Workmanship. The sonic prophylaxis unit shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One sonic prophylaxis unit, as specified, constitutes one unit.

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Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for inspection. Sampling for inspection shall be conducted in accordance with MIL-STD-105. Where feasible, the same sample shall be used for the determination of two or more inspection characteristics. The unit of product for inspection purposes shall be one prophylaxis unit. Unless otherwise specified, sampling for examination shall be conducted at inspection level II, AQL 2.5, and sampling for tests (when applicable) shall be conducted at inspection level S-1, AQL 1.0.

Examination. The unit (handpiece and tips) shall be examined to determine compliance with specification requirements.

Tests. Tests shall be conducted to determine compliance with specification requirements.

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Autoclave test. The unit shall be wrapped in muslin and autoclaved at 270 degrees F plus or minus 5 degrees F and 27 plus or minus 1 pounds per square inch for a minimum of 20 minutes. The unit shall then be inspected and tested for compliance with specification requirements.

Operational "run-in" test. Each unit shall be operated, "run-in", for 4 hours under normal operating conditions. The unit shall then be inspected and tested for compliance with specification requirements.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerance specified in the inch/pound units, a request should be made to the contracting officer to determine of the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

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Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document:

National Stock Number	Item Identification
6520-01-162-2906	SONIC PROPHYLAXIS UNIT, DENTAL, Portable, Air Driven

Ordering data. Intermediate/exterior package quantities and labeling and marking must be as specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

51793

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
A-A-53367A	SONIC PROPHYLAXIS UNIT, DENTAL, PORTABLE, AIR-Driven

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
3b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	7b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
7c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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